

On the Comparison of Dimensionality Reduction Techniques for News Classification

Wanaporn Chaungam, Pachalapa Yaicharaen, Punyanuch Chenyawanich, Junkaew Puandee

Department of Statistics, Faculty of Science, Kasetsart University, Bangkok 10900, Thailand

Abstract

Fact verification of news plays a crucial role in the modern era, as the rapid dissemination of fake news causes widespread negative impacts. Improving the efficiency of news classification in aspects of the execution time and the reduction in storage space is therefore important considerations. This study aims to compare the assistance of dimensionality reduction techniques, namely Principal Component Analysis (PCA), Locality Preserving Projection (LPP), Isometric Mapping (ISOMAP), and Laplacian Eigenmaps in fake news classification. In the dimensionality reduction of news, data are first transformed into vector representations using Sentence-BERT. The dimensionally reduced data are then classified using Random Forest. The performance is evaluated using 5-fold cross validation through Accuracy, Precision, Recall, and F1-Score as evaluation metrics. The results indicate that dimensionality reduced data accelerate the training time by more than 50% at an embedding dimension of 50, while preserving the classification performance comparable to that of the original data. These findings demonstrate that dimensionality reduction is an effective approach for efficient fake news classification.

Keywords: Dimensionality reduction technique, Fake news classification, Text embedding

Introduction

In the digital era, where information can be rapidly and widely distributed through social media platforms, the spread of fake news has become an increasingly serious problem. Such misleading information often causes confusion and may negatively affect society, the economy, and the mental well-being of individuals. Although various organizations and tools are available for fact-checking, many studies indicate that current systems are still unable to handle the large and continuously growing volume of online information effectively. Therefore, developing a reliable and efficient fake news classification system is essential.

Our approach begins with converting news text into vector representations using SBERT, which are then used as input for classification models. However, these vectors are typically high-dimensional and complex, making the classification process more challenging. As a result, dimensionality reduction is required, along with an assessment of the data's linearity, to ensure that an appropriate technique is selected.

This project examines several dimensionality reduction methods, including linear techniques such as PCA and LPP, as well as nonlinear techniques such as ISOMAP and Laplacian Eigenmaps. The dimensionally reduced data are subsequently classified using Random Forest to compare the effectiveness of each technique.

Since dimensionality reduction may distort the original structure of the data, evaluating the quality of the reduced representation is crucial. The Normalized Stress Function is used to measure how well each method preserves the relative distance structure of the original data. A threshold value of 0.15 is a common benchmark to determine whether the level of distortion is within an acceptable range.

To assess the classification performance, this project computes the average Accuracy, Precision, Recall, and F1-Score, combined with 5-fold cross-validation to ensure a reliable and well-rounded evaluation. Additionally, the average training and prediction time is documented to identify the most efficient technique in terms of computational resources.

In summary, this project aims to investigate and compare multiple dimensionality reduction techniques and apply the results to fake news classification, with a strong emphasis on evaluating each method's performance using appropriate and reliable metrics.

Equipment and methods

Equipment and program use in the process

A laptop was used to collect information and develop programs for data preprocessing, as well as to prepare documents for the project report. The programs used for data analysis and document preparation are Microsoft Excel, which is used for data cleaning, including checking the correctness of the data, removing data that is not needed for the experiment, and organizing the data into a suitable format for analysis. The project

report is prepared using Microsoft Word. In addition, the Python programming language is used for data processing and for building models for data analysis.

Method

In the dimensionality reduction stage, we employ both linear and nonlinear dimensionality reduction techniques. The linear techniques are PCA and LPP and the nonlinear techniques are ISOMAP and Laplacian Eigenmaps.

For all dimensionality reduction techniques, we set the embedding dimension ($n_components$) to 50, 100, 150, 200, and 227. Furthermore, we set the number of neighbors parameter ($n_neighbors$) to 5, 45, 85, 125, and 165 for LPP, ISOMAP, and Laplacian Eigenmaps. Using the same set of parameters allows a fair comparison of the performance of each technique.

After dimensionality reduction, the reduced data is used for news classification using Random Forest. The performance is evaluated using cross-validation. Although we carry out both linear and nonlinear dimensionality reduction techniques, they merely serves as a demonstration. The workflow in practical usage has been summarized in the following flowchart.

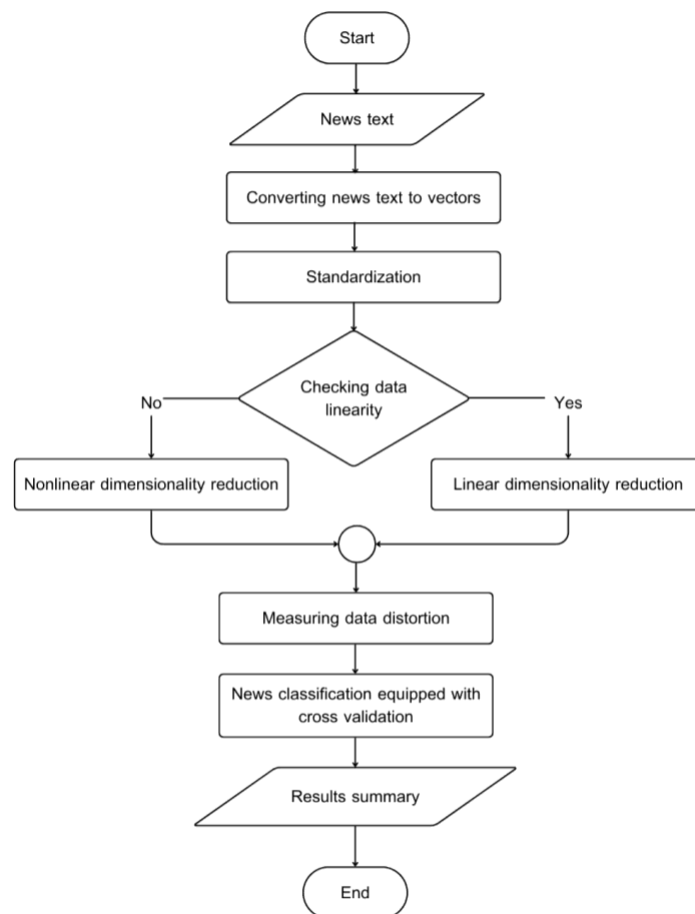


Fig. 1 Flowchart of dimensionality reduction techniques for news classification

Results and Discussion

This study evaluates the performance of dimensionality reduction techniques on the LIAR-DATASET (Wang, 2017). The dataset consists of 384-dimensional news vectors, transformed by Sentence-BERT. These vectors are then standardized to achieve a mean of 0 and a standard deviation of 1, to prevent the problem of scale differences among the features.

Then, the multicollinearity within the dataset was assessed by computing the covariance and correlation matrices to analyze the relationship patterns among the features prior to reduction. The analysis revealed that the positive covariance had a mean of 0.00016 with a maximum of 0.00134, while the negative covariance had a mean of -0.00016 with a minimum of -0.00127. However, covariance values depend on the scale of the variables and cannot directly indicate the strength of the relationships.

To provide a clearer assessment of the relationships among the features, the correlation matrix was examined, revealing that the positive correlations averaged 0.06772, with a maximum of 0.46746, whereas the negative correlations averaged -0.06928, with a minimum of -0.85056. These results indicate a wide distribution of correlations across the features. This trend can be observed in the heatmaps of the covariance and correlation matrices presented in Fig. 2.

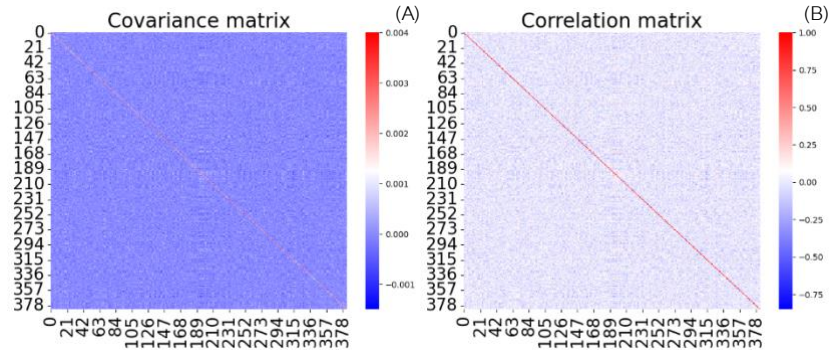


Fig. 2 Heatmap of covariance matrix (A); heatmap of correlation matrix (B)

From the analysis of the covariance and correlation matrices, we found that several features had a high level of correlation, reflecting feature redundancy within the dataset. These results are consistent with the VIF values calculated from the correlations among the features, with a mean of 7.93388 and a standard deviation of 2.23778. This indicates that the 384-dimensional vector data has high multicollinearity, which may affect the performance of the classification model before dimensionality reduction is applied.

Therefore, to study whether dimensionality reduction can be an approach to alleviate this problem, this study conducts experiments using PCA, LPP, ISOMAP, and Laplacian Eigenmaps embedding techniques before proceeding on to the data classification step.

Random Forest has been carried out for news classification by keeping the parameters `n_estimators`, `max_depth`, `min_samples_split`, and `random_state` constant for each technique to provide a fair comparison of dimensionality reduction performance, as shown in Table 1.

Table 1 Parameter settings for the Random Forest classifier

Parameter	Value
<code>n_estimators</code>	300
<code>max_depth</code>	10
<code>min_samples_split</code>	10
<code>random_state</code>	42

To make the performance evaluation more reliable, 5-fold cross validation was utilized. We divided the dataset into five equal subsets for iterative training and testing over five rounds. The obtained average performance metrics are then calculated to evaluate the models' performance.

To study the differences in news classification performance, this study conducts experiments under the same parameter configuration for each technique. The `n_components` parameter was considered for all techniques and `n_neighbors` was considered for LPP, ISOMAP, and Laplacian Eigenmaps. These parameters were adjusted to study their effects on classification performance. Furthermore, the original dataset without dimensionality reduction has been used as a baseline to compare the efficiency of each technique.

We show in Table 2 the average Accuracy, Precision, Recall, F1-score, and processing times when `n_neighbors` = 5 and `n_components` = 50. These parameters were chosen as a fixed configuration to evaluate all techniques under the exact same conditions.

Table 2 Performance comparison between Original Data and dimensionality reduction techniques reduced to 50 dimensions with `n_neighbors` = 5 for LPP, ISOMAP and Laplacian Eigenmaps.

Technique	Normalized Stress Function	Mean Accuracy	Mean Precision	Mean Recall	Mean F1-score	Mean Training Time (s)	Mean Testing Time (s)
Original Data	-	0.61227	0.61743	0.59357	0.58177	19.87705	0.05767
PCA	0.26079	0.61686	0.61981	0.59967	0.59062	7.99609	0.05314
LPP	0.93194	0.61758	0.62077	0.60050	0.59167	9.16963	0.06741
ISOMAP	3.69500	0.61347	0.61298	0.59844	0.59214	8.38420	0.05595
Laplacian Eigenmaps	0.99753	0.60526	0.60134	0.59358	0.59076	8.88484	0.06043

Particularly, Table 2 shows that the value that the Normalized Stress Function of the dimensionally reduced data using PCA and LPP, which are linear dimensionality reduction techniques, are significantly lower than these of ISOMAP and Laplacian Eigenmaps, which are nonlinear dimensionality reduction techniques. This result reflects the fact that the dataset used in the experiments has a linear relationship between dimensions, which is consistent with the examination of the multilinearity of the data.

Evidently, the Normalized Stress Function values in Table 2 emphasizes this difference more clearly. Specifically, the PCA and LPP techniques have values of 0.26079 and 0.93194, respectively, while the ISOMAP and Laplacian Eigenmaps techniques have values of 3.69500 and 0.99753, which are significantly higher than the PCA and LPP techniques. Particularly, in this dataset, linear dimensionality reduction techniques preserve the data structure better than nonlinear dimensionality reduction techniques.

Regardless of the embedding techniques, the dimensionally reduced data significantly decreases the training time, while Random Forest can still maintain classification performance at a level close to the original undimensionalized data. As shown in Table 2, PCA, LPP, ISOMAP, and Laplacian Eigenmaps reduce model training time by 59.77%, 53.87%, 57.82%, and 55.30%, respectively.

Summary

The four dimensional reduction techniques result in average Accuracy, Precision, Recall, and F1 scores that closely resemble the original data, although some techniques exhibit high levels of data distortion. This is because dimensional reduction may alter some of the internal structure of newsgroups, but the boundaries between genuine and fake news remain clearly defined, thus not affecting the model's news classification.

The key benefit of these four dimensionality reduction techniques is that they not only accelerate the speed of model training and news classification and reduce data storage space, but also maintain news classification performance close to that of the non-dimensionality reduction case. This demonstrates that dimensionality reduction does not decrease news classification performance; rather, it speeds up the process and uses fewer resources, making it a suitable approach for application in fake news classification.

Acknowledgements

This project, "On the Comparison of Dimensionality Reduction Techniques for News Classification," has been successfully completed. The project team would like to express our gratitude to Dr. Chanon Thongprayoon, our project advisor, for his invaluable support, guidance, and research assistance. He also helped review and correct any shortcomings, ultimately leading to the successful completion of this project. The team extends our thanks to our classmates for their assistance, exchange of ideas, and encouragement throughout the project. Finally, we extend our deepest gratitude to our parents and family for their unwavering support and encouragement at every stage of the process, which are crucial in the successful completion of

this project. Lastly, the project team hopes that this project will be beneficial to relevant organizations and those interested in further study.

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